Junior Makers · Class 6

Fire & Gas Safety Alarm Panel

A machine that never sleeps, so we can!

Infrared flame sensing | MQ-2 gas sensing | Alarm logic



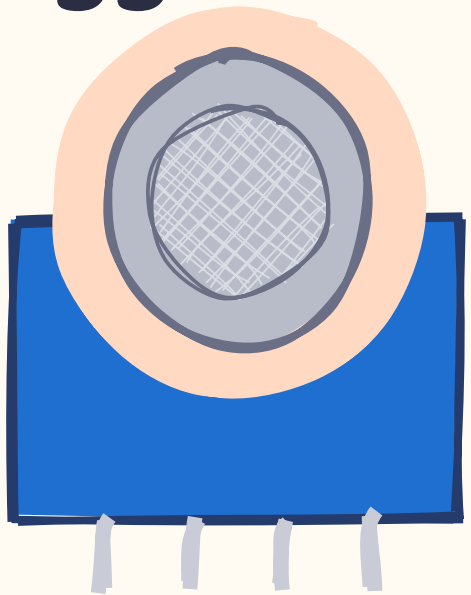
Did you know?
Most fire deaths happen at night, when people are asleep and cannot smell the smoke.

Let's think...

Why does Sui gas smell like rotten eggs?

A secret smell

Natural gas has **no smell at all**. Gas companies add a stinky chemical **on purpose**, so we can notice a leak.
But what if everyone at home is **asleep**? Or has a cold?



an electronic nose

Stay awake all night?

Get a dog?

Build an electronic nose!

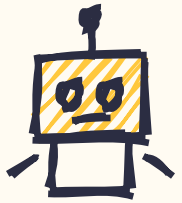
Today: we build a panel that sees fire, smells gas, and warns everyone in 3 different ways.

Learning aims

Today I will learn to...

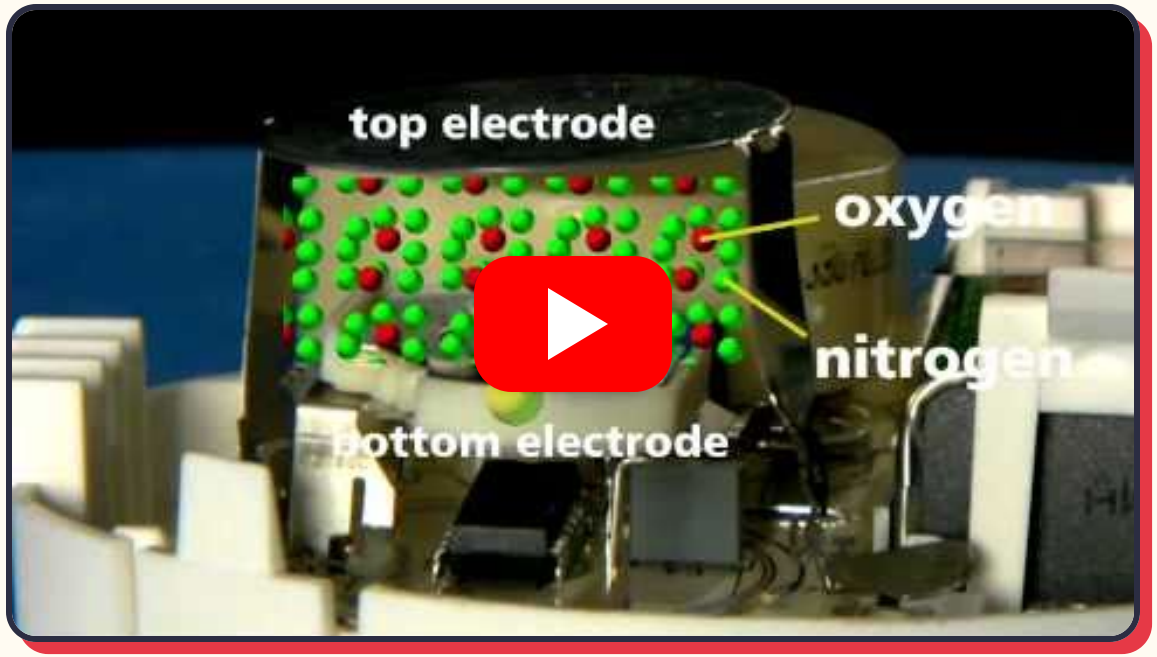
1. Explain the **fire triangle**: heat, fuel and oxygen
2. Use a **flame sensor** that sees invisible **infrared** light
3. Use an **MQ-2 gas sensor**, our electronic nose
4. Show **SAFE** or **DANGER** with lights, a buzzer and an OLED
5. Know exactly **what to do** in a real fire or gas leak

Keywords
fire triangle · infrared · gas · ppm · sensor · alarm · priority



Watch and think

Video time!



How Smoke Detectors Work

engineerguy · about 4 minutes · youtu.be/oFUUQcpGR3k

Watch for...

1. What does a smoke detector really **detect**?
2. Why does it make such a loud sound?
3. How often should you test a smoke alarm at home?

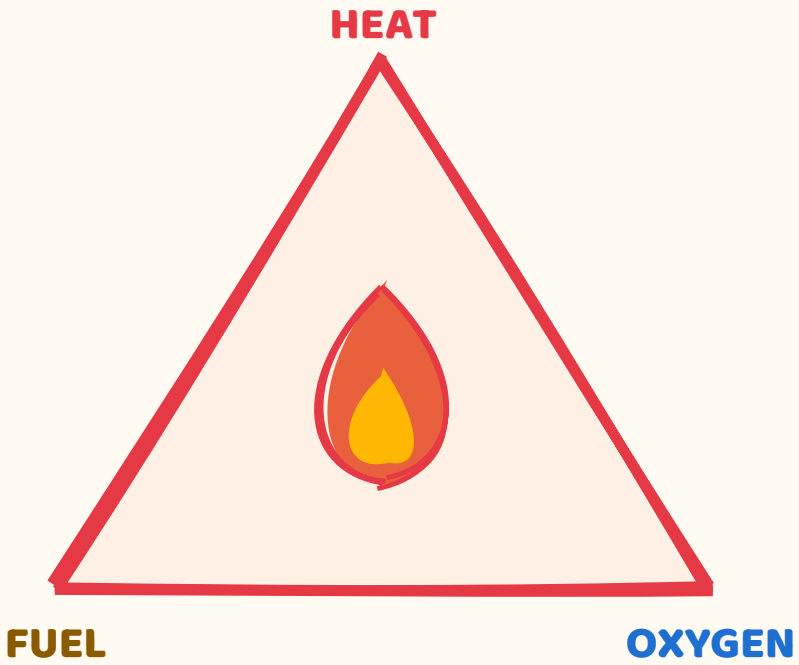
Scan to watch

After the video: With your partner, plan what your family would do if the alarm went off at night.

LEARN

The science

The fire triangle



Fire needs all three

Heat: a spark, a match, a hot stove

Fuel: wood, paper, cloth, **gas**

Oxygen: in the air all around us

Why gas is scary

Leaked gas is fuel floating in the air. One spark from a switch gives the heat. Boom!

Break the triangle!

Take away **any one** side and the fire goes out. A fire blanket takes away oxygen. Water takes away heat.

Try it (teacher)

Cover a small candle with a glass jar. Watch it go out. Which side did we remove?

Real world

Detectors that protect us



Smoke alarms
On ceilings in homes, schools and hotels



Kitchen gas alarms
Beep when cooking gas leaks from a cylinder or pipe



Factories
Shut off machines and call the fire brigade automatically



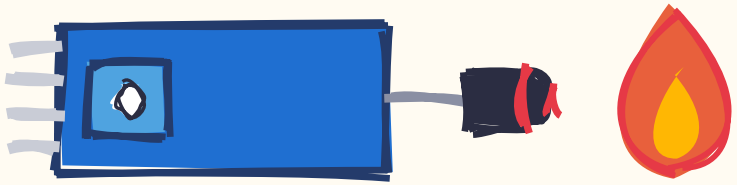
Aeroplanes
Fire sensors in every engine and in the toilets

? Look up at the ceiling. Can you spot a **smoke detector** in our school? How many in your home?

Winter warning
Gas heater and cylinder accidents happen most in winter. Alarms save lives.

Meet sensor 1

The flame sensor: it sees infrared



flame sensor



Flames give off **infrared** light, invisible to our eyes. The sensor's dark bulb (a **photodiode**) only sees infrared.

Pin	What it does
VCC	5V
GND	Ground
DO	LOW when it sees a flame
AO	Number: lower = stronger flame

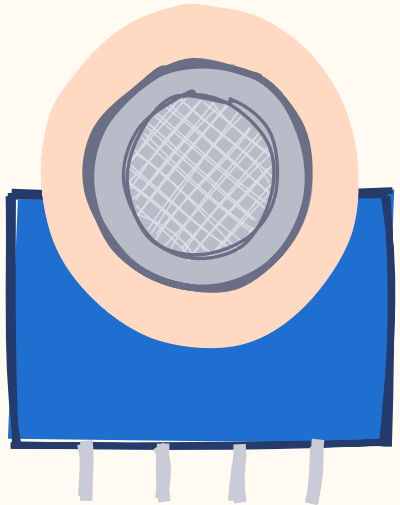
Fact file

Sees about **60°** wide · up to about **1 m** for a lighter flame · blue dial sets sensitivity

LEARN

Meet sensor 2

The MQ-2: an electronic nose



MQ-2 gas sensor

What it smells

LPG (cylinder gas), methane (Sui gas), butane (lighter gas), propane, hydrogen and **smoke**.

How it works

A tiny **heater** warms a special material. Gas changes how easily electricity flows through it. More gas = **bigger number** on AO.

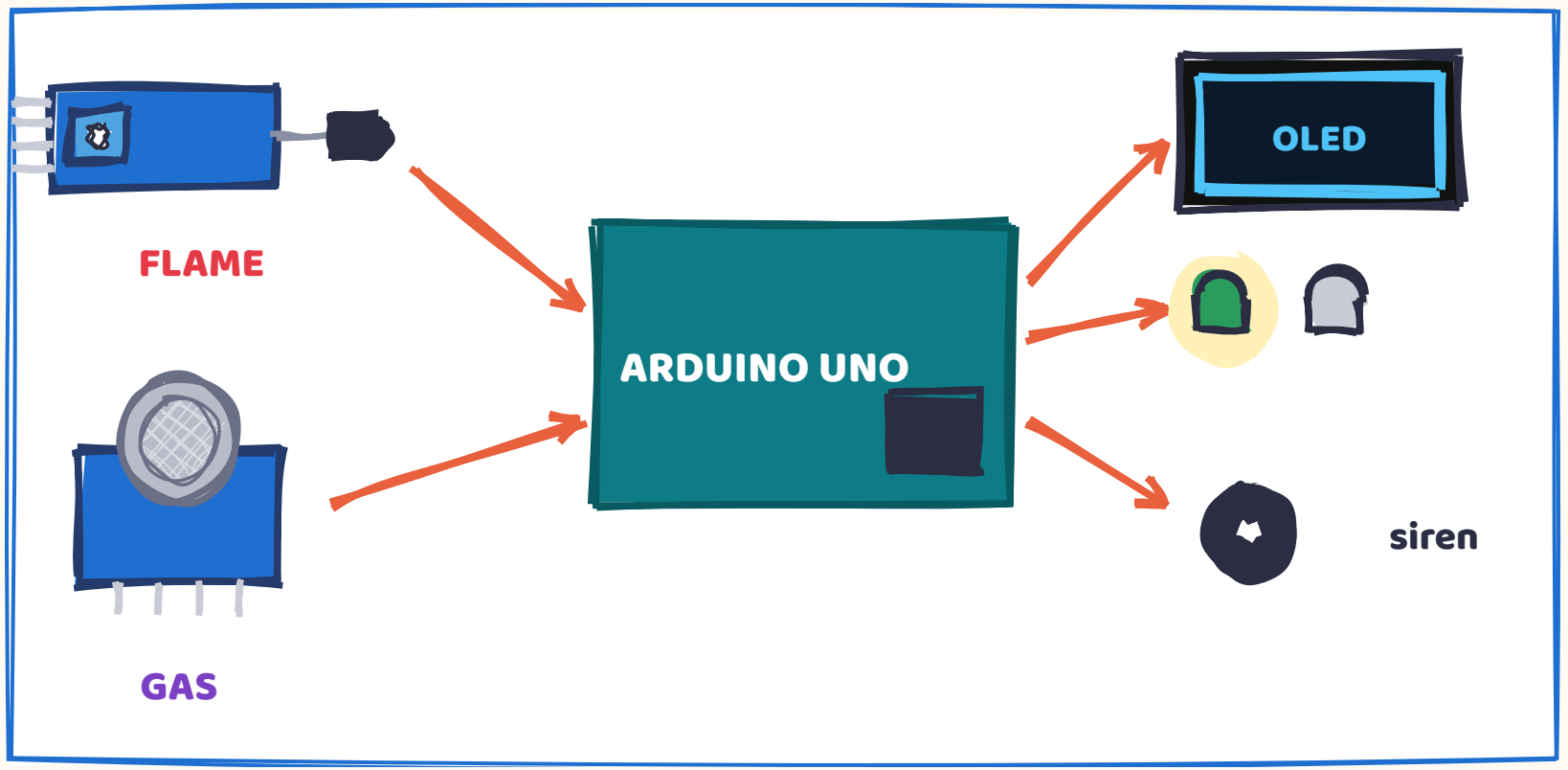
Hot! Hot! Hot!

The metal mesh gets **hot** when powered. Do not touch it!
It needs **1 to 3 minutes** to warm up before readings are steady.

Upside-down? Not this time!
Flame: lower = danger. Gas: HIGHER = danger. Engineers always check!

The plan

Two sensors, three warnings



Why 3 warnings?

Sound wakes people up.
Light helps people who cannot hear.
Words on the screen say **what** the danger is.

Design for everyone
A deaf grandparent would never hear a buzzer. A flashing light helps!

Life skills

If there is a real fire

1. **Shout "FIRE!"**
   warn everyone
2. **Get out fast**
   leave bags and toys
3. **Stay low**
   crawl under smoke, the air is cleaner
4. **Feel doors**
   hot door? use another way
5. **Stop, drop, roll**
   if your clothes catch fire
6. **Never go back**
   meet at the meeting point



Once you are **outside**, a grown-up calls **Rescue 1122** or the fire brigade on **16**.

Practise!
Act it out: everyone crawls low to the classroom door.

LEARN

Life skills

If you smell gas

- **Do NOT** switch lights or fans ON or OFF. A switch makes a tiny spark!
- **Do NOT** light a match, stove or lighter.
- **DO** open doors and windows to let the gas out.
- **DO** tell a grown-up to close the gas valve or cylinder.
- **DO** go outside, then call for help from there.



Remember why

Gas is the fuel. A spark is the heat. Do not complete the fire triangle!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Our safety panel kit (per group)
2. Wiring the safety panel
3. The safety panel in ArduinoBlocks
4. How we test: teacher only
5. Safety panel challenges
6. Real danger or false alarm?
7. Three states, three screens

Build it

Our safety panel kit (per group)



Arduino Uno
The brain



Flame sensor
Infrared eye



MQ-2 gas sensor
Electronic nose



OLED 0.96 in
The message screen



Green + red LED
With 2 × 220 Ω



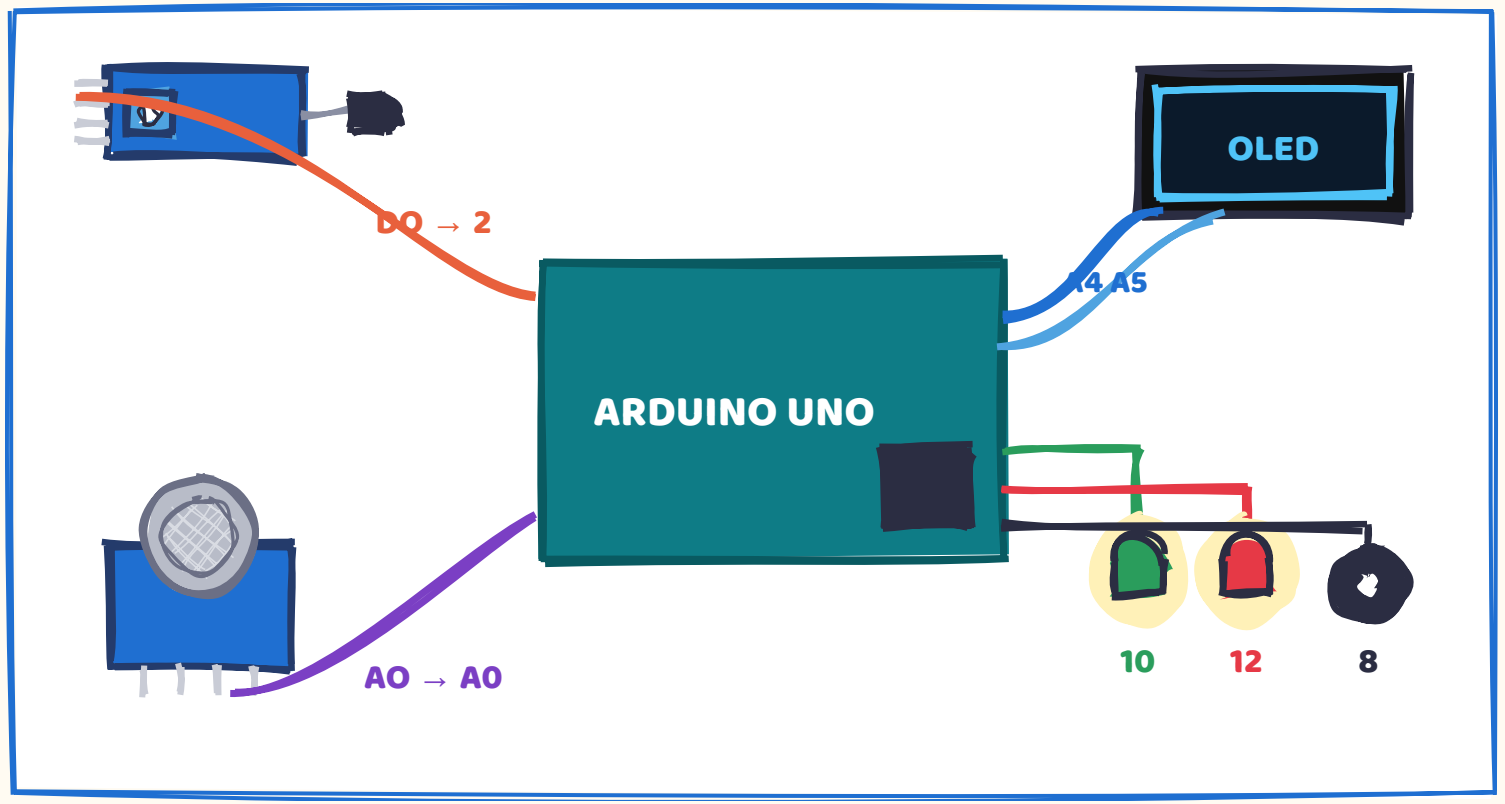
Buzzer
The siren

Also: breadboard, jumper wires, USB cable, laptop, card panel to mount it all.
Candle and lighter: teacher only.

Re-use
The OLED skills from the weather station come back today!

Build it

Wiring the safety panel



Part	Arduino
Flame VCC / GND	5V / GND
Flame DO	pin 2
MQ-2 VCC / GND	5V / GND
MQ-2 AO	A0
OLED SDA / SCL	A4 / A5
Green / red LED	10 / 12
Buzzer +	pin 8

Power tip
The MQ-2 heater is hungry. Plug the USB straight into the laptop, not a hub.

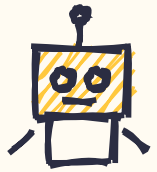
Code it

The safety panel in ArduinoBlocks

```
when Arduino starts
    set up OLED · wait 60 s (warm up)
forever
    set flame to read digital pin 2
    set gas to read analog A0
    clear OLED
    if flame = LOW
        red ON · green OFF · OLED 'FIRE!' · fast beeps
    else if gas > 400
        red ON · green OFF · OLED 'GAS LEAK' · slow beeps
    else
        green ON · red OFF · OLED 'SAFE' · quiet
    wait 200 ms
```

Read it like a story

Let the gas nose **warm up** first.
Check for **fire first** (most dangerous).
Else if gas is high: gas alarm.
Else: all safe, green light.



Find YOUR gas number
Print gas to the Serial monitor in clean air. Set the threshold about 150 above that number.

Safety first

How we test: teacher only

Fire test

1. Candle on a **metal tray**, far from paper and hair.
2. Teacher holds it about **30 cm** from the flame sensor.
3. Children watch from **behind the line**.
4. Blow out. Panel goes back to SAFE?

Gas test

1. **Blow out the candle first.** No flame anywhere!
2. Teacher presses an **unlit** lighter for **1 second** near the MQ-2.
3. Watch the number rise, then fall.
4. Open a window afterwards.

Golden rule: never do the gas test and the fire test at the same time. Gas + flame = real danger!

MAKE

Level up!

Safety panel challenges

Choose your challenge



Flash for everyone: make the red LED flash during an alarm, for people who cannot hear.



Silence button: add a button that stops the siren but keeps the light flashing.



Gas meter: show the gas number as a bar on the OLED.



Two sirens: a different sound for fire and for gas, so you know which one without looking.

Think like an engineer

Should a real alarm stop by itself when the danger goes away? Or keep warning until a person checks?

Boss level

Only sound the gas alarm if gas stays high for 3 readings in a row. Fewer false alarms!

Team game

Real danger or false alarm?

For each card, decide: **which sensor would react?** Is it **real danger** or a **false alarm**?

1 Candle in the dark	**2** Bright sunshine through the window
3 Gas cylinder valve left open	**4** Burnt toast smoke
5 TV remote pointed at it	**6** Perfume spray

Answers (teacher): 1 flame · 2 flame, false alarm (sunlight has infrared!) · 3 gas, real danger · 4 gas, maybe real · 5 flame, false alarm (remotes use infrared) · 6 gas, false alarm (perfume has alcohol)

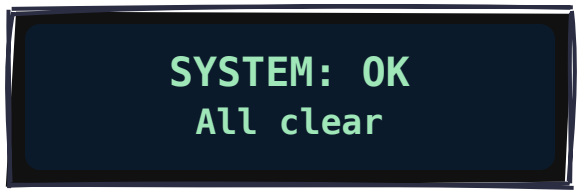
Big idea

Every sensor can be fooled. Good engineers test for false alarms too!

Design the panel

Three states, three screens

SAFE



Green ON · red OFF · quiet

FIRE!



Red ON · fast siren

GAS LEAK!



Red ON · slow beeps

Priority: what if there is fire AND gas? Check **fire first**. The most dangerous message wins the screen!

Your turn

Design your own message. Keep it short: the screen is tiny!

Test like an engineer

Does every alarm work?

Test	Expect	Gas number	Result
Normal room air	SAFE, green	_____	pass / fail
Candle at 30 cm (teacher)	FIRE!, fast siren	_____	pass / fail
Lighter gas, unlit (teacher)	GAS LEAK, slow beeps	_____	pass / fail
Torch / phone light	SAFE? (false alarm test)	_____	pass / fail
Sunlight on the sensor	?? Try it!	_____	pass / fail

Debug checklist

Always FIRE? Turn the blue dial on the flame sensor.
Always GAS? Wait for warm-up, or raise the threshold.
Blank OLED? Check A4 / A5 and the address (0x3C).

Print **flame** and **gas** on the Serial monitor. Numbers tell the truth!

Show what you know

What did we learn?

- [x] Fire needs heat, fuel and oxygen
- [x] A flame sensor sees invisible infrared light
- [x] The MQ-2 nose gives a bigger number when there is more gas
- [x] Check the most dangerous thing first: priority
- [x] Gas leak: no switches, open windows, get out

Exit ticket: why should you NOT switch on a light if you smell gas?

At home

With a grown-up, find where the gas valve or cylinder is. Plan your family's meeting point outside.



Next class: a Temperature-Controlled Smart Fan!